POCI — Three Recovery Scenarios

Demo script · PCS Loan Module V4 · IFRS 9 §5.5.13–14, §B5.4.7

Figures regenerated 20 October 2026 against the engine, after the credit-adjusted EIR solve was corrected. Every number below is engine output, not an independent calculation. See "What changed on 20 October" at the end.

Before you start — read this

One thing is still broken, and it is the last journal. The engine's maturity leg repays **face value** unconditionally. On Scenario 1 that is right. On Scenarios 2 and 3 it is not — the app will post 100,000,000 when only 78,000,000 or 60,000,000 is actually received, overstating the gain.

This is open issue **#71**. Run the demo through the **Cashflow** and **Carrying value** views, where every figure below is now correct, and stop short of the maturity journal unless you want to discuss the issue openly. If a client asks, the honest answer is: *"the amortised-cost mechanics are right; the final principal leg currently assumes full recovery and we're fixing it."*

What is no longer broken: until 20 October the credit-adjusted EIR was solved on an annual cash-flow approximation that ignored coupon frequency, and the carrying amount overshot the expected recovery by up to 3%. Both are fixed. If you have an older copy of this script, discard it — its figures will not match the screen.

The set-up — identical in every respect but one

All three deals are the same purchase:

Face value	**100,000,000 USD**
Consideration paid	**60,000,000 USD** on 1 Jan 2026
Coupon	**8.00% fixed**, quarterly, cash, ACT/360, on **face**
Maturity	31 Dec 2031 · bullet · 2,191 days
Classification	Amortised cost · credit-impaired at acquisition · **POCI**

One input differs: the recovery expected at acquisition. That single number sets the credit-adjusted EIR, and everything else follows from it. Keeping all other terms identical is deliberate — anything a client sees differing between the three screens is caused by the expected recovery and nothing else.

Deal code	Expected recovery	Lifetime ECL at acquisition	Credit-adjusted EIR
`DEMO-POCI`	75,000,000	25,000,000	**15.7803%**
`DEMO-POCI-MID`	78,000,000	22,000,000	**16.2241%**
`DEMO-POCI-PAR`	60,000,000	40,000,000	**13.3333%**

The coupon is 8.00% in all three. No EIR is anywhere near it. That is the first thing to say out loud, and it is the whole of POCI in one line: the coupon tells you what the borrower pays; the EIR tells you what the investment earns. They are different numbers answering different questions.

The rate convention, before anyone asks

These are **nominal ACT/360 simple** rates — the basis the schedule accrues on, so they are the rates that actually drive the carrying amount. A client who takes the cash flows and runs an annual-effective IRR will get a higher number. Both are right; they are different conventions. Have the equivalents ready:

Deal	Nominal ACT/360 (shown)	Annual effective
`DEMO-POCI`	15.7803%	17.3464%
`DEMO-POCI-MID`	16.2241%	17.8754%
`DEMO-POCI-PAR`	13.3333%	14.4720%

Why the EIR is so far above the coupon

You pay 60m for a claim on 100m. The coupon is 8% **on the face**, so every quarter you collect roughly 2.0m of cash on a 60m investment — about 13% a year before any principal recovery at all. Add whatever principal comes back above the 60m you paid and the return rises further. The credit-adjusted EIR is the single rate that discounts the **expected** cash flows — contractual flows less expected credit losses — back to the price paid (§B5.4.7).

Two consequences worth stating:

1. **No day-one allowance.** The losses are already in the price. A POCI asset opens at 60,000,000 with a nil allowance, where a Stage 1 asset bought at par would open with an allowance against it (§5.5.13).
2. **No stages.** POCI assets never enter the 1/2/3 staging model. If the screen shows "Stage 1" on these deals it is a display default, not a measurement — that is open issue **#72**.

Scenario 3 first — `DEMO-POCI-PAR`

Run this one first. It is counter-intuitive and it makes the other two obvious.

Expected recovery 60,000,000 — exactly what you paid.

Q1 2026:

EIR income at 13.3333%	2,000,000.00
Cash coupon received	2,000,000.00
Accretion	**0.00**
Carrying value	60,000,000.00

Carrying value at each year end: 60,000,000.00 — the same figure, six times. Not approximately. Exactly.

Say this: *"There is no accretion here, and that is correct. Accretion is the pull of the carrying amount towards what you expect to collect. If you expect to collect exactly what you paid, there is nothing to pull towards. The entire return on this deal is the coupon — 108.7m collected over six years on a 60m outlay, 1.81×."*

Then the point that lands: **the EIR is still 13.33%, not 8%.** And this one you can check in your head on the spot:

13.3333% = 8.00% × 100,000,000 / 60,000,000

The coupon is 8% of face; you only paid 60% of face; so the yield on what you actually invested is the coupon grossed up by 100/60. The engine solves it numerically and lands on exactly that. A client who books interest income at the coupon rate on the carrying amount would recognise roughly 4.8m a year instead of 8.1m.

This deal is the single best integrity check in the pack. Flat carrying, nil accretion, and an EIR you can verify by mental arithmetic. If any of the three is off, something is wrong upstream.

Scenario 2 — `DEMO-POCI-MID`

Expected recovery 78,000,000 — between the 60m paid and the 100m face.

Q1 2026:

EIR income at 16.2241%	2,442,432.33
Cash coupon received	2,000,000.00
Accretion	**442,432.33**
Carrying value	60,442,432.33

Carrying value path, at each year end:

2026	61,911,008.89	accreted to date 1,911,008.89
2027	64,163,617.56	4,163,617.56
2028	66,826,779.27	6,826,779.27
2029	69,958,099.74	9,958,099.74
2030	73,649,155.10	13,649,155.10
2031	**78,000,000.00**	**18,000,000.00**

Say this: *"Now there is something to pull towards. You paid 60m and expect 78m, so 18m of the return is principal recovery rather than coupon — and IFRS 9 makes you recognise it over the life, as interest income, not as a windfall at the end. The carrying amount climbs from 60m to exactly 78m on the maturity date. It cannot overshoot: the EIR was solved to land it there."*

That last sentence is now literally true and worth proving. Open the **Unaccreted** and **Accreted to date** columns on the Cashflow tab side by side. At any period, *discount still to take + discount already taken = the day-one discount of 18,000,000*:

	accreted to date	unaccreted	sum
end 2026	1,911,008.89	16,088,991.11	18,000,000.00
end 2029	9,958,099.74	8,041,900.26	18,000,000.00
end 2031	18,000,000.00	0.00	18,000,000.00

Year 1 totals: EIR income 10,022,120.00 · cash coupon 8,111,111.11 · accretion 1,911,008.89.

Scenario 1 — `DEMO-POCI`

Expected recovery 75,000,000 at acquisition. The estimate is then revised upward in 2028, and the borrower ultimately pays the full 100,000,000.

This one is about what happens when reality beats the assumption — and it has **two** favourable events, not one. The earlier version of this script only described the second.

Credit-adjusted EIR	15.7803%
Gross carrying value at maturity	75,000,000.00

Year 1: EIR income 9,725,015.22 · cash coupon 8,111,111.11 · accretion 1,613,904.11.

Gross carrying path: 61,613,904.11 → 63,507,762.30 → 65,736,726.58 → 68,345,745.59 → 71,407,335.18 → **75,000,000.00**

The 2028 revision — this is the part to demo

The deal carries a stored revision dated **1 January 2028**: lifetime expected loss cut from 25,000,000 to 15,000,000. Watch the three columns across that date:

	gross carrying	ECL allowance	**net carrying**
31 Dec 2027	63,507,762.30	0.00	63,507,762.30
1 Jan 2028	63,513,378.23	**(10,000,000.00)**	**73,513,378.23**
31 Dec 2031	75,000,000.00	(10,000,000.00)	85,000,000.00

Say this: *"The allowance has gone negative, and net carrying is now higher than gross. On any other asset that is impossible. Under §5.5.14 a favourable change in expected cash flows on a POCI asset is recognised as an impairment **gain** — and critically, it is recognised even though it takes the loss allowance below nil, because the comparison is against the expected flows at acquisition, not against zero."*

Two things to point at while that is on screen:

1. **The EIR did not move.** It is 15.7803% before the revision and 15.7803% after. The gross carrying path is identical either side. **The EIR is never revised for a POCI asset** — that is what "credit-adjusted" buys you (§B5.4.7). The revision lands entirely in the allowance.
2. **The gain is 10,000,000 in 2028**, not at maturity. If the borrower then pays the full 100,000,000 against a net carrying amount of 85,000,000, the remaining gain at maturity is **15,000,000** — not 25,000,000. The total favourable swing is 25,000,000; it arrives in two pieces, and the demo shows both.

This is also the one scenario where the engine's maturity leg is right — it posts face, and face is what was received.

The three side by side

	Scenario 3 PAR	Scenario 2 MID	Scenario 1 POCI
Paid	60,000,000	60,000,000	60,000,000
Expected at acquisition	60,000,000	78,000,000	75,000,000
Credit-adjusted EIR	13.3333%	16.2241%	15.7803%
Coupon	8.00%	8.00%	8.00%
Gross carrying, end 2026	60,000,000.00	61,911,008.89	61,613,904.11
Gross carrying at maturity	60,000,000.00	78,000,000.00	75,000,000.00
Accretion over life	0.00	18,000,000.00	15,000,000.00
Total coupon over life	48,688,888.89	48,688,888.89	48,688,888.89
ECL allowance at maturity	—	—	(10,000,000.00)
On expected recovery			
Total collected	108,688,888.89	126,688,888.89	123,688,888.89
Multiple on cost	1.81×	2.11×	2.06×

	Scenario 3 PAR	Scenario 2 MID	Scenario 1 POCI
If the borrower pays face			
Total collected	—	—	148,688,888.89
Multiple on cost	—	—	2.48×

Note Scenario 1's EIR (15.78%) is **lower** than Scenario 2's (16.22%) despite collecting far more cash. That is not a mistake and it is a good question to invite: the EIR is set at acquisition from what you *expected*, and Scenario 1 expected 75m against Scenario 2's 78m. The extra arrives as a gain, not as yield.

Questions you will be asked

"Why is interest income higher than the cash we received?" Because the discount is income. You bought at 60m something you expect to collect more on; the difference accrues as interest over the life rather than appearing as a gain at the end. Scenario 2 is the clearest: 10.0m of income in 2026 against 8.1m of cash.

"Your EIR doesn't match my IRR." Check the convention before checking the arithmetic. The screen shows a nominal ACT/360 simple rate because that is what the schedule accrues on. The annual effective equivalents are in the table at the top — 17.88% for Scenario 2 against the 16.22% on screen.

"What if the expected recovery changes mid-life?" Re-discount the revised expected flows at the **original** credit-adjusted EIR. The change in carrying amount is an impairment gain or loss immediately. The EIR itself never moves. Scenario 1 does exactly this on 1 Jan 2028.

"Can the allowance go negative?" On a POCI asset, yes — and Scenario 1 shows it: (10,000,000) from 2028, with net carrying above gross. A favourable revision is recognised as a gain even if cumulative favourable changes exceed the losses expected at acquisition (§5.5.14). On any non-POCI asset, no.

"Why is there no allowance on day one?" Because the expected losses are already in the 60m you paid. Recognising an allowance as well would count them twice (§5.5.13).

"Is this Stage 3?" No. POCI sits outside the staging model entirely. If the screen shows a stage, that is a display default — issue #72.

"What happens to a POCI asset that cures completely?" It stays POCI for life. The classification is fixed at acquisition and never changes, however the credit improves (§5.5.13).

Checklist before you present

- ☐ All three deals load and show a **credit-adjusted EIR, not the coupon**
- ☐ `DEMO-POCI-PAR` shows **13.3333%** — check it: 8% × 100/60
- ☐ `DEMO-POCI-PAR` carrying is **60,000,000.00 flat**, nil accretion
- ☐ `DEMO-POCI-MID` carrying lands on **78,000,000.00 exactly** at maturity
- ☐ On MID, accreted-to-date + unaccreted = **18,000,000** at every period
- ☐ Day-one allowance is **nil** on all three
- ☐ Gross carrying amount opens at **60,000,000**, not 100,000,000
- ☐ On `DEMO-POCI`, the allowance goes to **(10,000,000)** on 1 Jan 2028 and the EIR does **not** move
- ☐ **Do not run to the maturity journal on MID or PAR** — issue #71
- ☐ Have the issue numbers to hand: #71 maturity leg, #72 stage label

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What changed on 20 October 2026

The previous version of this script carried figures computed independently of the engine. They were the correct IFRS 9 answer; the engine did not reproduce them. Checking the two against each other found a real defect, now fixed.

The defect. The POCI branch of the yield solve built its expected cash-flow vector as one payment a year and never read the coupon frequency, so a quarterly payer was solved as if it paid annually in arrears. It also discounted at an annual effective rate while the schedule accrues daily and simple on ACT/360 — two different conventions. The result was an EIR understated by about 100bp and, more seriously, a carrying amount that **did not converge on the expected recovery**: `DEMO-POCI-MID` finished at 80,379,504.83 against a target of 78,000,000, an overshoot of 3.05%.

The fix. The annual vector is now only a seed. It is refined against the real daily walk until the carrying amount lands on the expected recovery, which is the same treatment the ordinary `effectiveInterestPrice` path already had. All three deals now converge to **0.00** drift. Both engine harnesses (25 + 25 cases) pass unchanged.

Two documentation errors fixed at the same time, independent of the engine:

- Scenario 1's 2028 ECL revision was in the deal data but not in the script. The gain is 10,000,000 in 2028 plus 15,000,000 at maturity, not 25,000,000 at maturity.
- The old side-by-side table mixed expected and actual recovery in one column for Scenario 1. They are now separate blocks.

Issues **#71** and **#72** are unchanged and still open.